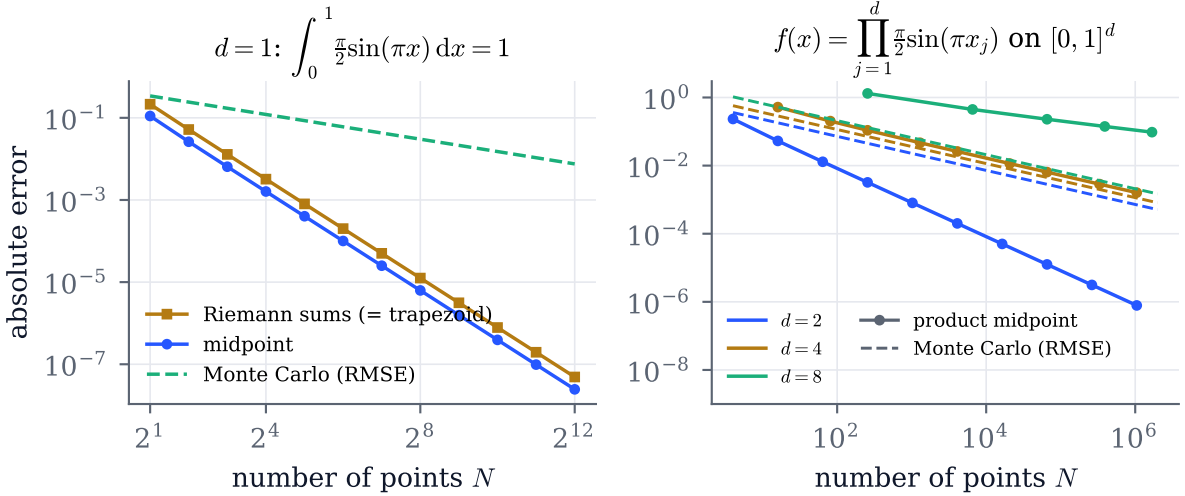
$d = 1$: $\int_0^1 \frac{\pi}{2}\sin(\pi x)\,\mathrm{d}x = 1$
absolute error
10^{-1}
10^{-3}
10^{-5}
10^{-7}
Riemann sums (= trapezoid)
midpoint
Monte Carlo (RMSE)
2^1
2^4
2^8
2^{12}
number of points N
$f(x) = \prod_{j=1}^{d} \frac{\pi}{2}\sin(\pi x_j)$ on $[0,1]^d$
10^0
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$d = 2$
$d = 4$
$d = 8$
product midpoint
Monte Carlo (RMSE)
10^2
10^4
10^6
number of points N